

BRIEFING FOR SOCIAL HOUSING PROVIDERS

Warm, affordable, zero-carbon homes — with minimal disruption

How the Hestora thermal energy store helps social landlords decarbonise hard-to-treat stock, protect tenants from fuel poverty, and make retrofit funding go further.

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1. The challenge you're being asked to solve

Social landlords are being asked to do three difficult things at once: bring every home to EPC C by 2030, decarbonise heating on the road to net zero, and protect tenants from fuel poverty — all across stock that includes solid walls, tight terraces, flats over shops and homes with no outside space. Heat pumps are the default answer, but for a large slice of the stock they carry real delivery pain: outdoor units, emitter and cylinder upgrades, fabric prerequisites, multi-day installs, tenant decants and a new heating experience that generates call-outs.

Hestora is designed for precisely that slice: a like-for-like boiler replacement powered by cheap off-peak electricity, installed in around a day, using the radiators and pipework already in the property.



Installation takes around a day — no outdoor unit, no new radiators — sited in a utility space, garage, airing cupboard or loft.

2. What Hestora is

Hestora is a 40 kWh pressurised, water-based thermal energy store. It charges from off-peak electricity in a four-to-five-hour overnight window and delivers a full day of space

heating and mains-pressure hot water — behaving, from the tenant's point of view, exactly like the boiler it replaces. The unit occupies the space of a hot-water cylinder — 650 mm × 650 mm and 1,800 mm tall — and because it arrives light and dry it can be sited almost anywhere in the property: garage, utility room, airing cupboard or loft. Water held under pressure at up to 175 °C gives it four to five times the energy density of a conventional cylinder; round-trip efficiency is ~92%; the pressure vessel is independently certified to PD 5500:2024+A2 (Category 2) and tested to more than double its operating pressure by British Engineering Services. Design life is over 30 years — water, steel and insulation, with no refrigerants and no battery cells.

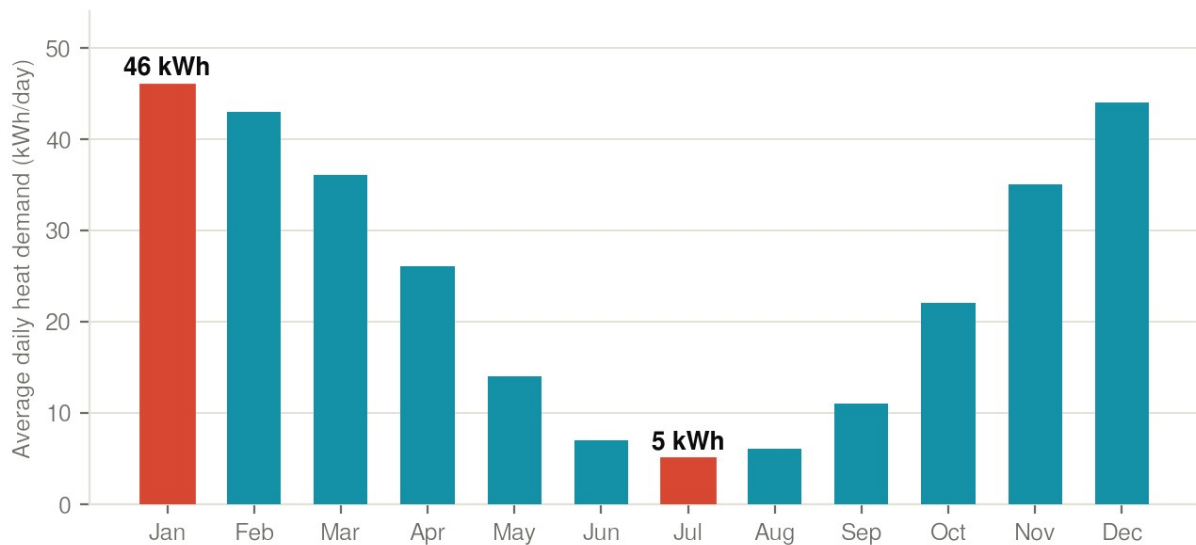
3. Why it fits social housing

- **Minimal disruption** — installation takes around a day, sited wherever suits the property — garage, utility, airing cupboard or loft. No outdoor unit, no emitter upsizing, no fabric prerequisite, no scaffolding. Tenants keep their heating routine; landlords keep their programme moving.
- **Works in hard-to-treat homes** — terraces without outside space, conservation-constrained properties and poorly insulated homes that heat pumps struggle to serve. Initial focus: houses, bungalows and maisonettes with wet central heating.
- **Tenant acceptance built in** — same radiators, same controls logic, mains-pressure hot water, boiler-like responsiveness. Flexibility happens invisibly in the background — no behaviour change to explain, fewer call-outs to field.
- **Fuel-poverty protection** — the household's largest energy load moves to ~7p/kWh overnight electricity, and consumption is decoupled from winter-evening price spikes. For landlords managing fuel-poverty risk, predictability matters as much as the saving — it is the extreme bill spikes that do the damage.
- **Portfolio economics** — a >30-year design life spreads capital across decades and avoids the 10–15-year replacement cycle of more complex technologies — a materially stronger NPV for asset-management plans.
- **A future income stream** — Phase 1 charging uses the overnight off-peak window; from mid-2027, units will additionally integrate with NESO's Demand Flexibility Service 'turn-up' mechanism, which pays controllable demand to absorb surplus renewable power (the participation threshold fell to 0.1 MW in April 2026, within reach of even a small fleet). Each unit can absorb 40 kWh on command — aggregated across a portfolio, a genuine dispatchable asset.
- **Solar-ready** — where stock has (or gains) rooftop PV, a built-in diverter charges the store from the roof before the grid. Depending on array size, that can mean virtually no heating and hot-water costs for the tenant from April to October.

4. What our modelling shows

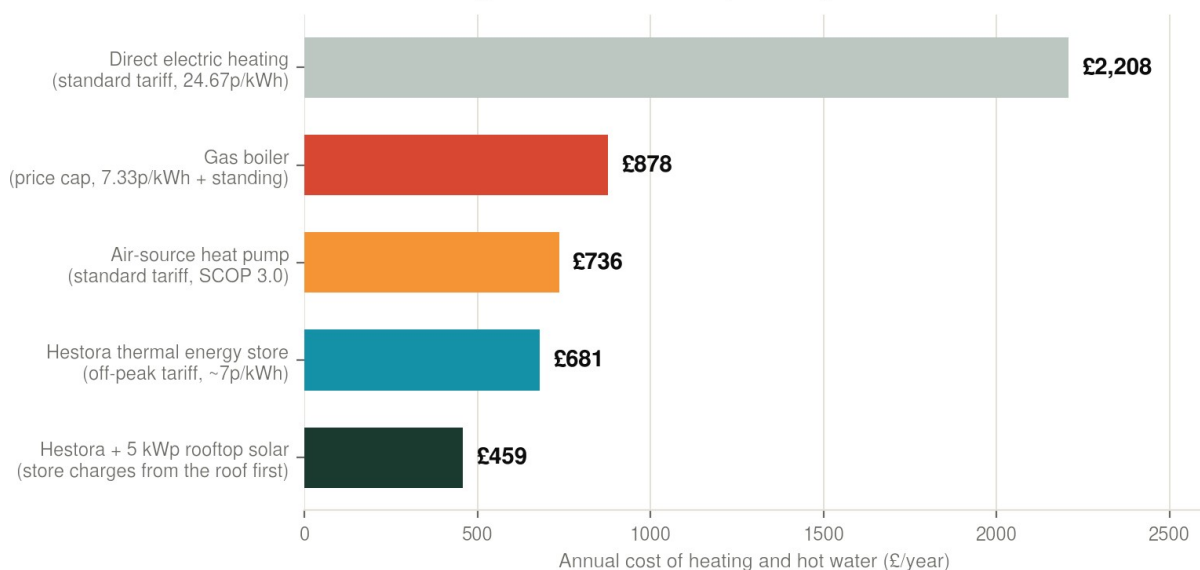
Hestora's sizing and economics are grounded in primary data: a living-lab analysis of a full year of half-hourly smart-meter readings — about 6.4 million data points — from 200 British homes.

Heat demand of a typical UK gas-heated home across the year



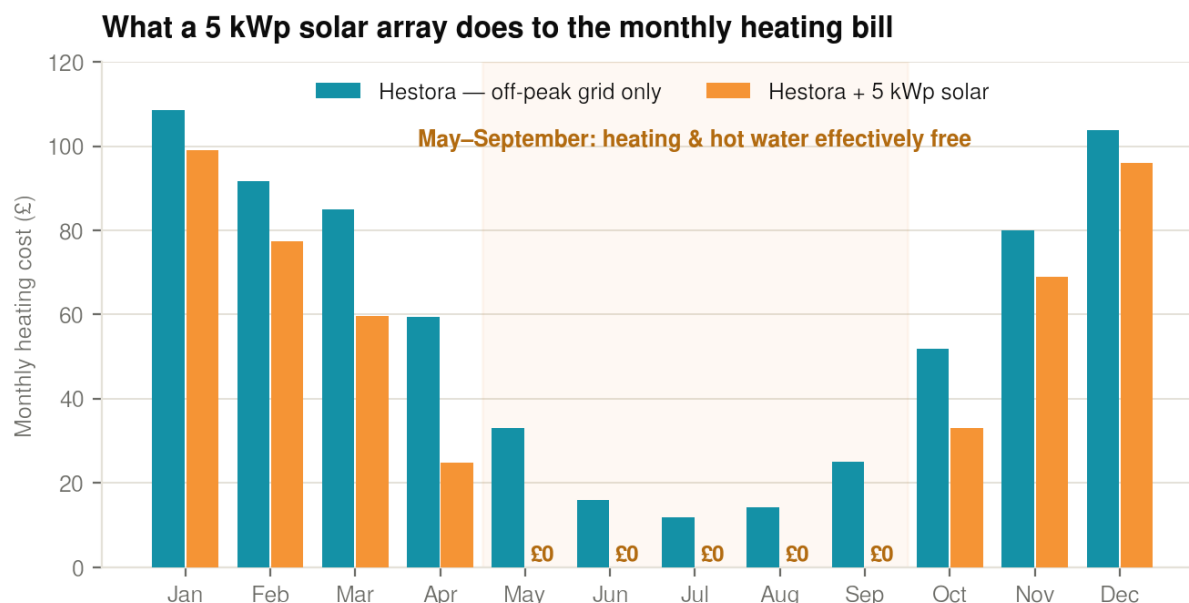
Average daily heat demand by month for a typical gas-heated home (~8,950 kWh/yr), from the living-lab dataset of 200 homes (~6.4 million half-hourly readings). A single 40 kWh overnight charge covers the maximum daily demand of 47% of the UK stock; on the deepest winter days the control system tops up in secondary off-peak windows.

Annual running cost — same home, five ways to heat it



Modelled annual cost of meeting 8,950 kWh of heat and hot-water demand, mid-2026 prices. Assumptions: direct electric at the standard single rate (24.67p/kWh) — the position much electrically-heated social stock is already in; gas at the Jul-Sep 2026 cap (7.33p/kWh plus standing charge, 85% boiler efficiency); air-source heat pump at SCOP 3.0, spanning £631–£788 across the plausible 3.5–2.8 range; Hestora at ~7p/kWh off-peak, and with a 5 kWp array feeding the store first. For tenants on legacy storage or panel heating, the move to either of the bottom two bars transforms the bill.

Solar is fast becoming a key asset on social housing roofs — and Hestora turns it directly into lower tenant bills. With the store charging from the roof before the grid, a 5 kWp array makes heating and hot water effectively free from May to September:



Monthly heating and hot-water cost with Hestora charged from off-peak grid electricity alone (blue) and with a 5 kWp south-facing array feeding the store ahead of the grid (amber). May–September falls to zero and the annual cost drops from ~£681 to ~£459, the array supplying roughly a third of annual charging energy. Monthly-resolution model against the living-lab demand profile; a larger array widens the zero-cost window.

5. Funding and policy alignment

- **Warm Homes: Social Housing Fund** — £1.29bn allocated for 2025–28 (Wave 3), within a wider commitment of £13.2bn of retrofit funding to 2030, with works completing by September 2028. Funding is devolved in the West Midlands and Greater Manchester. The Wave 3 guidance already anticipates our category: thermal energy stores become eligible once MCS-certified, installed by MCS-certified installers using products listed in the MCS Product Directory. A measure installed in around a day helps committed funds convert into completed homes before the deadline.
- **Boiler Upgrade Scheme** — access to the £2,500 BUS grant depends on the BSI standard for thermal energy storage being defined, alongside MCS certification and testing by BRE — like other thermal energy stores, Hestora is waiting for these to be confirmed. Note the two routes do not stack: under the Wave 3 rules, a property that has received BUS funding is not eligible for WH:SHF low-carbon heating funding, so landlords choose the route that suits each property.
- **The Future Homes Standard** — from March 2027 new-build submissions in England must cut emissions ~75–80% against the 2013 baseline, with gas effectively ruled out and rooftop solar expected on most new homes. For new social housing development, Hestora pairs naturally with that mandated solar: the store soaks up the roof's surplus generation as well as cheap overnight power, and adds no load to the evening peak — which also eases network-connection negotiations on larger sites.

Worked example — net cost per property (BUS route)

£7,000 installed – £2,500 BUS grant – ~£1,000 carbon-offset funding = **~£3,500 net**, once the BSI standard, MCS certification and BRE testing are confirmed. Alternatively, WH:SHF programme funding can cover the measure (the two routes cannot be combined on the same property). Either way, a >30-year life makes the cost per warm, zero-emission year a fraction of shorter-lived alternatives.

6. An invitation: pilot with us

A monitored, instrumented pilot begins in November 2026, ahead of commercial sales in 2027. We are looking for social landlord partners who want early access to the technology, tenant-experience data from real homes over the winter.

Next steps

Visit www.hestora.net to find more information, or contact jim@hestora.net to discuss a pilot in your stock. Hestora Limited is a British engineering company based in the Midlands, supported by energy and enterprise programmes at Imperial College London and the University of Birmingham.